Transformer Model for Functional Near-Infrared Spectroscopy Classification

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***Abstract*—Functional near-infrared spectroscopy (fNIRS) is a promising neuroimaging technology. The fNIRS classification problem has always been the focus of the brain-computer interface (BCI). Inspired by the success of Transformer based on self-attention mechanism in the fields of natural language processing and computer vision, we propose an fNIRS classification network based on Transformer, named fNIRS-T. We explore the spatial-level and channel-level representation of fNIRS signals to improve data utilization and network representation capacity. Besides, a preprocessing module, which consists of one-dimensional average pooling and layer normalization, is designed to replace filtering and baseline correction of data preprocessing. It makes fNIRS-T an end-to-end network, called fNIRS-PreT. Compared with traditional machine learning classifiers, convolutional neural network (CNN), and long short-term memory (LSTM), the proposed models obtain the best accuracy on three open-access datasets. Specifically, in the most extensive ternary classification task (30 subjects) that includes three types of overt movements, fNIRS-T, CNN, and LSTM obtain 75.49%, 72.89%, and 61.94% on test sets, respectively. Compared to traditional classifiers, fNIRS-T is at least 27.41% higher than statistical features and 6.79% higher than well-designed features. In the individual subject experiment of the ternary classification task, fNIRS-T achieves an average subject accuracy of 78.22% and surpasses CNN and LSTM by a large margin of +4.75% and +11.33%. fNIRS-PreT using raw data also achieves competitive performance to fNIRS-T. Therefore, the proposed models improve the performance of fNIRS-based BCI significantly.**

***Index Terms*—Brain-computer interface (BCI), classification, end-to-end, functional near-infrared spectroscopy (fNIRS), preprocessing module, self-attention, transformer.**

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I. Introduction

FUNCTIONAL near-infrared spectroscopy (fNIRS) is a new non-invasive neuroimaging technique to detect the hemodynamic responses of brain activities by measuring the absorption of the near-infrared light between 650 and 950 nm [1]. It determines the concentration changes of oxygenated hemoglobin (HbO) and deoxygenated hemoglobin (HbR) from diffusely scattered light measurements [2]. Many other neuroimaging techniques including electroencephalography (EEG), magnetoencephalography (MEG), and functional magnetic resonance imaging (fMRI) are also employed for brain signal analysis. EEG uses metal electrodes and conductive media to record electrical signals generated by neurons. Compared with fNIRS signals, EEG has a better time resolution, but its spatial resolution is lower, and it is more vulnerable to electrical noise and motion artifacts [3]. Although fMRI can provide excellent spatial resolution [4], fMRI signal acquisition is a time-consuming process and will produce a lot of noise. More importantly, fMRI equipment is bulky and expensive, so it is not suitable for the mobile scene. Considering security, portability, and relatively low cost [3], fNIRS has more advantages. Therefore, fNIRS signal analysis has attracted the attention of researchers in recent years.

How to efficiently use fNIRS signals to classify different tasks has always been the focus of research. The classic fNIRS classification process mainly includes data acquisition, data preprocessing, feature engineering, and training classifiers. Traditional machine learning classifiers such as linear discriminant analysis (LDA), k-nearest neighbor (k-NN), support vector machine (SVM), and artificial neural network (ANN) are widely used in the brain-computer interface (BCI). SVM was utilized to obtain the best performance in schizophrenia classification [5]. LDA achieved an average classification accuracy of $86.5 \pm 6.0\%$ on 9 subjects [6]. A five-layer multi-layer perceptron (MLP) was applied to classify motor imagery tasks for hybrid EEG-fNIRS BCI [7]. Traditional machine learning methods use hand-crafted features for classification, which depend on the prior knowledge and experience of experts. The subjectivity of feature engineering limits the classification accuracy and generalization ability of BCI systems. In recent years, deep neural networks (DNNs) have achieved great success in the field of computer vision (CV) [8], [9] and natural language processing (NLP) [10], [11]. Deep learning methods can automatically extract features and learn complex information from input data, which include convolutional neural networks (CNNs), recurrent neural networks (RNNs), and long short-term memory (LSTM). In classifying discrete emotions induced by affective video film clips, deep learning models combined with complexity markers obtained superior performance [12]. The average accuracy of LSTM was 89.31% in the four-class cognitive states classification, which was much higher than SVM, k-NN, and ANN [13]. In four-class motor imagery classification tasks, CNN achieved an average classification accuracy of 72.35 $\pm$ 4.4% [14].

Different from image data, the representation of the fNIRS signal is also a critical procedure. To improve data utilization and network representation capacity, we explore two different levels of data representation. fNIRS signals usually contain multiple channels in which every single-channel signal is time-series data. The multi-channel fNIRS signal can be treated as image data since it has image characteristics and forms multiple matrices. We apply the spatial-level representation that can cover multiple channels and the channel-level representation to detect the hemodynamic response of a single channel. In the deep learning community, convolutional layers are used to extract image features [15], and LSTM utilizes memory cells and gate units to process sequence information [16]. However, CNNs can only focus on local neighborhoods, and LSTM is difficult to capture long-distance context, which causes many optimization difficulties. Therefore, a new neural network with global context and parallel computing arouses our interest.

Transformer [10], a new type of deep model based on self-attention mechanism, is initially applied in the NLP field and obtains state-of-the-art results. Devlin *et al.* [11] proposed a model called BERT (Bidirectional Encoder Representations from Transformers), which refreshed the record of eleven NLP tasks. Brown *et al.* [17] trained a GPT-3 (Generative Pre-trained Transformer 3) with 175 billion parameters and achieved excellent performance in downstream tasks. Due to the powerful representation ability of Transformers, they have made significant progress in the CV tasks. Vision Transformer (ViT) [18] is the first model that uses pure Transformer to replace standard CNNs on large-scale image datasets. The great success of Transformers in two different fields inspires us to use this model to solve fNIRS classification problems. The main contributions of this paper are as follows:

1. To the best of our knowledge, there is no research on the application of Transformers for fNIRS classification tasks. We are the first to propose an fNIRS classification network based on Transformer, named fNIRS-T.

2. We explore the spatial-level and channel-level representation of fNIRS signals to improve data utilization and network representation capacity. Moreover, to extract low-level features and promote optimization procedures, convolutional inductive biases are introduced to replace the patch embedding of ViT.

3. Few researchers have studied using deep learning modules to replace filtering and baseline correction of fNIRS data preprocessing. We design a preprocessing module that is composed of one-dimensional (1D) average pooling and layer normalization [19], which makes fNIRS-T an end-to-end classification network, called fNIRS-PreT.

4. To solve overfitting and improve generalization, we use label smoothing [20] instead of the cross-entropy loss function and adopt the latest flooding [21] regularization method.

5. Extensive experiments on three open-access datasets prove that the proposed models significantly outperform traditional classifiers, CNN, and LSTM. Visualization and ablation studies further illustrate superior performance.

II. Materials

A. Dataset Acquisition

Dataset A[1] is to study the changes of HbO and HbR in the prefrontal cortex during mental arithmetic (MA) and to classify single-trial data stably and reliably [22], [23]. Since this study does not design the baseline (BL) task, the rest period is labeled as REST by previous studies [22], [24].

Dataset B[2] provides an open-access dataset to study the visual instruction of MA tasks [25]. We introduce MA and BL tasks for classification experiments, and the dataset is more complex than Dataset A.

Dataset C[3] is an open-access dataset for three-class classification to facilitate fNIRS studies [26]. The dataset contains three types of overt movements: right-hand finger-tapping (RHT), left-hand finger-tapping (LHT), and foot-tapping (FT). Unlike Dataset A and B, we classify the three movements instead of BL or REST. Therefore, this dataset is more challenging than other datasets.

Table I lists the number of subjects, experimental paradigms, and task specifications for each dataset.

B. Data Preprocessing

The modified Beer-Lambert law [27] is applied to convert optical density ΔOD at time Δt into the concentration changes of deoxygenated hemoglobin (ΔHbR) and oxygenated hemoglobin (ΔHbO) from the absorption of near-infrared light. The law is described as

$$\begin{bmatrix} \Delta HbR \\ \Delta HbO \end{bmatrix} = \frac{1}{d}\begin{bmatrix} \varepsilon_{HbR}\left(\lambda_1\right) & \varepsilon_{HbO}\left(\lambda_1\right) \\ \varepsilon_{HbR}\left(\lambda_2\right) & \varepsilon_{HbO}\left(\lambda_2\right) \end{bmatrix}^{-1} \begin{bmatrix} \Delta OD\left(\Delta t, \lambda_1\right)/DPF\left(\lambda_1\right) \\ \Delta OD\left(\Delta t, \lambda_2\right)/DPF\left(\lambda_2\right) \end{bmatrix}, \tag{1}$$

where d is the distance between source and detector, λ_1 and λ_2 are the different illumination wavelengths, DPF is the differential path length factor of λ, ε is the extinction coefficient of HbR and HbO. A simple filter is applied to remove physiological noise and motion artifacts. Dataset A uses a 0.09 Hz

[1][Online]. Available: http://bnci-horizon-2020.eu/database/data-sets

[2][Online]. Available: http://doc.ml.tu-berlin.de/hBCI

[3][Online]. Available: https://doi.org/10.6084/m9.figshare.9783755.v1

TABLE I
DETAILS OF THE THREE DATASETS

Dataset	Subject	Paradigm	Task specification
Dataset A	8 (3 males, 5 females, 26.0 ± 2.8 years old)	Mental arithmetic; Introduction period (2 s); Task period (12 s); Rest period (28 s)	For the instruction period, the participants were required to memorize an initial subtraction displayed on the screen. During the task period, they were asked to sequentially calculate to subtract a one-digit number from the previous subtraction result. During the rest period, they were instructed to take a break by gazing at the black screen.
Dataset B	29 (14 males, 15 females, 28.5 ± 3.7 years old)	Mental arithmetic; Introduction period (2 s); Task period (10 s); Rest period (15–17 s)	For the MA task, it is similar to Dataset A. For the BL task, these subjects were allowed to relax and gazed at the appearing black cross. At the beginning and end of the task period, a short beep (250 ms) was played.
Dataset C	30 (17 males, 13 females, 23.4 ± 2.5 years old)	Motor execution; Introduction period (2 s); Task period (10 s); Rest period (17–19 s)	The assigned task was initiated with a short beep, the volunteers were required to continuously perform a movement according to the instruction on the screen, including RHT, LHT, and FT. The rest period started with a short beep and a "STOP" sign on the screen, and they were allowed to relax.

low-pass Butterworth filter of order 4 with 60 dB [22]. Dataset B utilizes a band-pass filter with a passband of 0.01–0.1 Hz [25]. Dataset C applies a third-order Butterworth filter with a passband of 0.01–0.1 Hz [26]. Baseline correction is used to deal with baseline drift. For the preprocessing methods, we follow the previous literature. Finally, fNIRS signals are divided into small fragments for classification. Due to the delayed characteristic of brain hemodynamic responses [28], the divided fragments contain the delayed task-related parts.

C. Feature Extraction

The classification performance is highly dependent on feature extraction. The mean, variance, peak value, slope, skewness, and kurtosis of statistical features are widely used for classification tasks [3], [29], [30]. These statistical features are normalized in the range of 0 and 1 using min-max normalization to accelerate the optimization process. It is written as

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)}, \tag{2}$$

where x is the raw feature values, x' is the normalized feature values, $\max(x)$ and $\min(x)$ are the maximum and minimum values of x, respectively. These statistical features are used to train machine learning classifiers.

III. METHOD

A. Spatial-Level and Channel-Level Representation

How to reasonably use fNIRS signals is a key issue related to classification results. Convolutional inductive biases, such as translation invariance and locality, are introduced into Transformers to extract low-level features and promote optimization procedures [31], [32]. The spatial-level and channel-level are two different methods to improve data utilization and network representation capacity. The relationship between the two-level representations is shown in Fig. 1. The fNIRS matrix is expressed as an image $x \in R^{H \times W \times C}$, where H is the number of channels, W is the number of sampling points, and C is two chromophores (HbO and HbR). This arrangement can better maintain temporal and spatial information. Then, fNIRS signals are normalized by the z-score standardization. The spatial-level representation can obtain multiple channel information and focus on different activation areas via convolution $Conv_S$. The channel-level

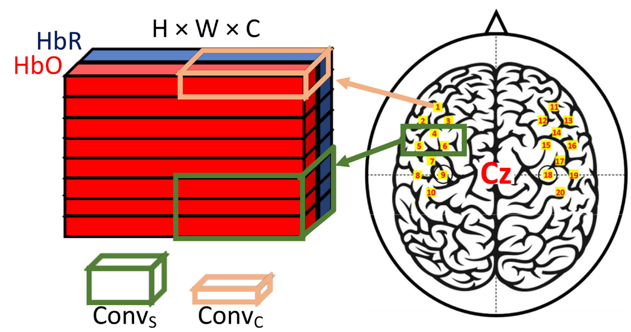


Fig. 1. The relationship between the spatial-level and channel-level representation. The picture of fNIRS channel locations is from [26].

representation can pay close attention to the brain hemodynamic response of a single channel via convolution $Conv_C$. Transformers based on self-attention mechanism assign different attention weights to brain activation areas and hemodynamic responses. The combination of the two-level representations makes up for the shortcomings of a single method.

B. fNIRS-T and fNIRS-PreT Architecture

Transformer is a new network structure based on self-attention mechanism, and its amazing performance in the NLP field has attracted significant attention in the CV field. The feature extraction of Transformers is different from CNNs and RNNs. Transformers can easily focus on global information, while CNNs can only extract local information. Compared with RNNs, Transformers can efficiently deal with longer sequences and parallelized calculations [10]. Therefore, we propose an fNIRS classification network based on Transformer, called fNIRS-T. The model is composed of two parts: fNIRS Spatial-level Transformer (fNIRS-ST) and fNIRS Channel-level Transformer (fNIRS-CT). fNIRS-ST can extract local brain area features through multiple channels but ignores the hemodynamic response of a single channel. fNIRS-CT can pay more attention to the hemodynamic response of each channel. Besides, we design a simple preprocessing module to replace the filtering and baseline correction of data preprocessing. The preprocessing module is integrated with fNIRS-T, which makes fNIRS-T become an end-to-end classification network, named fNIRS-PreT. Fig. 2 shows the structure of our proposed models. Code

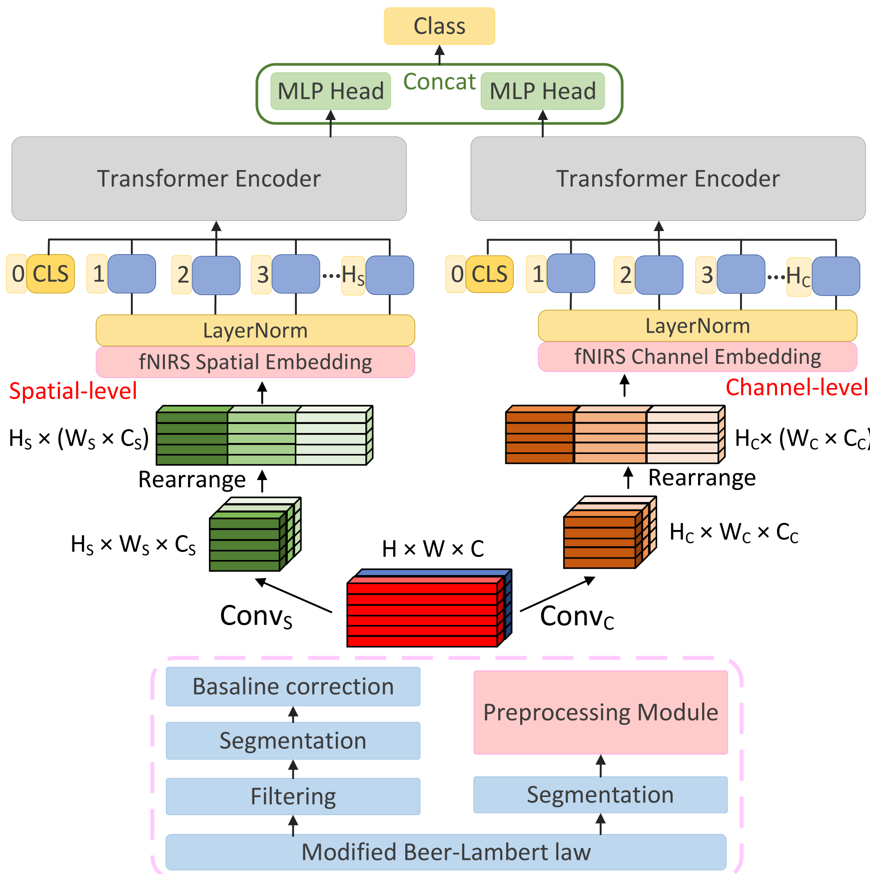


Fig. 2. Overview of the proposed model architecture. fNIRS-T is composed of fNIRS-ST (left) and fNIRS-CT (right). The bottom part is the classic data preprocessing. The preprocessing module is utilized to replace the filtering and baseline correction of data preprocessing.

TABLE II
PARAMETER SETTINGS OF THE CONVOLUTIONS

Convolution	Input size	Size / Filter / Stride	Output size
$Conv_S$	$H \times W \times C$	(5, 30) / 8 / (1, 4)	$H_S \times W_S \times C_S$
$Conv_C$	$H \times W \times C$	(1, 30) / 8 / (1, 4)	$H_C \times W_C \times C_C$

will be available at https://github.com/wzhlearning/fNIRS-Transformer.

First of all, the fNIRS matrix x must be converted into tokens in the NLP application. The patch operation of ViT is implemented by a stride-p $p \times p$ convolution ($p = 16$), which runs counter to the typical design of CNNs and brings some optimization difficulties [31]. We introduce inductive biases into the spatial-level and channel-level representation by convolution $Conv_S$ and $Conv_C$ to improve data representation and network optimization. The details of the convolutions are listed in Table II. The output feature map sizes of $Conv_S$ and $Conv_C$ are $H_S \times W_S \times C_S$ and $H_C \times W_C \times C_C$, respectively. Then, the feature maps are rearranged along the W axis to maintain fNIRS channel location information and facilitate separation into small sequences. At last, the two feature maps are divided into H_S sequences $x_S \in R^{1\times(W_S\times C_S)}$ and H_C sequences $x_C \in R^{1\times(W_C\times C_C)}$, respectively. These sequences are projected to D dimension vector by using a linear embeddings, and then trainable 1D position embeddings are attached to retain positional information. It is worth noting that a learnable classification token $[CLS]$ is appended at the beginning of the spatial and channel embeddings. The token is similar to BERT. After layer normalization [19], these joint embeddings serve as input to Transformer encoders. The final MLP head corresponding to the $[CLS]$ is used as the aggregate sequence representation and the two representation parts of the MLP head are concatenated into one feature vector for classification.

C. Transformer Encoder Structure

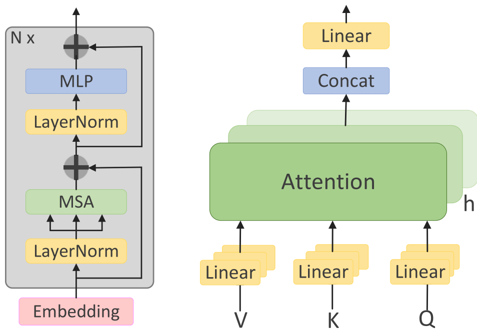
Fig. 3(a) shows the structure of Transformer encoder. It consists of identical layers, and each layer has two sub-layers: a multi-head self-attention (MSA) and an MLP. As shown in Fig. 3(b), the input vector of the self-attention layer is firstly transformed into queries Q, keys K, and values V. The self-attention is defined as

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \quad (3)$$

where d_k is the dimension of K. MSA is composed of h self-attention layers in parallel, then we have

$$MSA(Q, K, V) = Concat\,(head_1, \ldots, head_h)\,W^O, \quad (4)$$

$$head_i = Attention\left(QW_i^Q, KW_i^K, VW_i^V\right), \quad (5)$$



(a) (b)

Fig. 3. (a) and (b) are the structure of Transformer encoder and MSA, respectively.

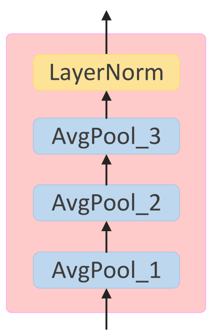


Fig. 4. Preprocessing module structure.

where W_i^Q, W_i^K, W_i^V, and W^O are parameter matrices. The MLP consists two linear layers with a Gaussian error linear unit (GELU) [33] activation function. It is defined as

$$MLP(x) = GELU\left(xW_1 + b_1\right)W_2 + b_2, \tag{6}$$

where W_1 and W_2 are weights of the two fully-connected layers, b_1 and b_2 are the bias terms. Layer normalization is a crucial method to stabilize training and accelerate convergence [19]. It is performed before MSA and MLP rather than after, and this change can train Transformers more efficiently [34]. Besides, a residual connection [8] is added in each sub-layer to strengthen information flow and avoid network degradation.

D. Preprocessing Module

Whether a traditional machine learning or a deep learning method is employed, fNIRS signals preprocessing seems to be an inevitable part. The preprocessing steps usually include the modified Beer-Lambert law, filtering, and baseline correction. It is necessary for classification to segment signals. However, few researchers have studied how to use deep learning modules to replace these preprocessing steps directly. Therefore, we design a preprocessing module, only using 1D average pooling and layer normalization to replace signal filtering and baseline correction, respectively. The modified Beer-Lambert law which converts the raw optical density signals into hemoglobin concentration changes has a clear physical meaning and cannot be omitted. The specific information is shown in Fig. 4 and Table III. The HbO and HbR channels of fNIRS matrices will be preprocessed by this module.

TABLE III
PARAMETER SETTINGS OF THE PREPROCESSING MODULE

Layer	Kernel size	Stride	Padding
$AvgPool\_1$	5	1	2
$AvgPool\_2$	13	1	6
$AvgPool\_3$	7	1	3
$LayerNorm$	-	-	-

Average pooling is regarded as a moving average filter that is widely used for noise removal in signal processing [35]–[37]. We can change the size of the sliding filter window by adjusting the average pooling kernel size. For 1D average pooling, a smaller kernel can retain the main information and avoid over-averaging. In contrast, a larger kernel is used to smooth signals and remove noise. In the preprocessing module, the first 1D average pooling has a smaller kernel, the second has a larger kernel, and the third is mainly for fine smoothing. In Table III, we set the padding parameters to make the input and the output data have the same size, which allows the preprocessing module to be easily integrated with fNIRS-T.

The normal baseline correction is to subtract the mean value of the reference interval μ_0 from the fNIRS signal sample x, which is written as

$$BC(x) = x - \mu_0. \tag{7}$$

This baseline correction only performs simple shift operations on the signal. Layer normalization is written as

$$LayerNorm(x) = \frac{x-\mu}{\sigma} \odot g + b, \tag{8}$$

where μ is the mean, σ is the standard deviation, $\odot$ is the element-wise multiplication, and g and b are defined as the gain and bias parameters, respectively. Layer normalization is rewritten as

$$LayerNorm(x) = \frac{g}{\sigma} \odot x - \left(\frac{\mu \odot g}{\sigma} - b\right), \tag{9}$$

where g and b are obtained through neural network training. Layer normalization can scale and shift signals and has a similar function to baseline correction. Moreover, it can also accelerate convergence and reduce training time.

E. Training Strategy

1) Label Smoothing: Label smoothing is initially proposed by Szegedy *et al.* [20], and this strategy is used to improve the classification performance of the Inception model. When using deep networks to train fNIRS data, overfitting that affects classification results is prone to occur. Label smoothing can prevent the model from becoming over-confidence and improve generalization [38]. The probability of each label $k \in \{1 \ldots K\}$ is first calculated as

$$p_k = \frac{\exp\left(z_k\right)}{\sum_{i=1}^{K} \exp\left(z_i\right)}, \tag{10}$$

TABLE IV
KEY CONFIGURES OF OUR MODEL VARIANTS

Dataset	Layers	Hidden dim	MLP dim	Heads	Params
Dataset A	6	64	64	8	1.7M
Dataset B	6	64	64	8	1.7M
Dataset C	6	128	64	8	3.5M

where z_i is the logits or unnormalized log-probabilities. Then, the cross-entropy loss function is written as

$$L(y,p) = \sum_{k=1}^{K} -y_k \log(p_k), \tag{11}$$

where y_k is "1" for the correct class and "0" for the rest. Finally, y_k is replaced by smoothed label y_k^{LS}. It is defined as

$$y_k^{LS} = (1-\alpha)\, y_k + \frac{\alpha}{K}, \tag{12}$$

where the smoothing parameter α is usually set to 0.1. The label smoothing is written as

$$L\left(y^{LS}, p\right) = \sum_{k=1}^{K} -y_k^{LS} \log(p_k). \tag{13}$$

2) Flooding: Flooding [21] is a surprisingly simple regularization method to avoid overfitting and improve test accuracy. When the overfitting happens, the training loss is decreasing and the test loss is increasing. Flooding can keep the training loss around a small value called flooding level to avoid zero training loss and reduce the test loss. The label smoothing $L(y^{LS}, p)$ with flooding is

$$\tilde{L}\left(y^{LS}, p\right) = \left|L\left(y^{LS}, p\right) - b\right| + b, \tag{14}$$

where b is the flooding level. In experiments, we decay flooding levels at specific training epochs to obtain continuous performance improvement.

IV. RESULTS

A. Baseline Traditional Classifiers and DNNs

In this study, traditional classifiers and DNNs are used to conduct comparison experiments with our models. Traditional classifiers include k-NN, ANN, and SVM. The k-NN parameter k is set to 50. ANN has 128 hidden layer units, and the maximum iteration is 10000. We use SVM with the radial basis function kernel, and the regularization parameter C is set to 1. To compare with DNNs, we utilize fNIRS classification networks proposed by other researchers. The CNN model [39] consists of three 1D convolutional layers, two fully connected layers, and a softmax layer. Each 1D convolutional layer has 32 filters with a kernel size of 3. The LSTM model [13] is composed of four LSTM layers and two fully connected layers. Each LSTM layer has 64 hidden units, and the fully connected layers contain 512 and 128 hidden nodes, respectively.

B. Training Parameters

Our model variables are summarized in Table IV. Models are trained for 120 epochs with a batch size of 128, using AdamW [40] with an initial learning rate of 0.001, decay parameters $\beta_1 = 0.9$ and $\beta_2 = 0.999$, and a weight decay of 0.01. We utilize label smoothing and cosine learning rate decay [41] to accelerate training and improve performance. The cosine learning rate can accelerate the convergence during the decay phase and surpass poor local optima during the restart phase. It is worth noting that the restart phase often temporarily worsens performance, but it is beneficial in the long term [41]. In addition, we decay flooding levels at specific epochs to obtain a continuous regularization effect. Flooding levels for deep networks are listed in Table V. Different networks may use different values. Before training these models, flooding levels are determined by optimal test results. We conduct 5 runs $\times$ 5-fold cross-validation to evaluate the average classification accuracy of the test sets. Finally, to comprehensively evaluate individual subject classification performance, a leave-one-subject-out cross-validation (LOSO-CV) strategy is employed to evaluate the generalization ability [42]. In this strategy, one subject's data is used as the test set, and the rest is used as the training set. Repeat this process until all subject data has been tested. We further select precision, recall, F1-score (macro F1-score for Dataset C), and Kappa coefficient to illustrate subject classification results.

TABLE V
FLOODING LEVELS FOR DEEP NETWORKS

Dataset	Method	Flooding[1]
Dataset A	CNN	-
	LSTM	(1, 0.50), (30, 0.45), (50, 0.40)
	fNIRS-T	-
	fNIRS-PreT	-
Dataset B	CNN	(1, 0.55), (30, 0.50), (50, 0.45)
	LSTM	(1, 0.65), (30, 0.60), (50, 0.55)
	fNIRS-T	(1, 0.45), (30, 0.40), (50, 0.35)
	fNIRS-PreT	(1, 0.40), (30, 0.38), (50, 0.35)
Dataset C	CNN	(1, 0.50), (30, 0.45), (50, 0.40)
	LSTM	(1, 0.70), (30, 0.65), (50, 0.60)
	fNIRS-T	(1, 0.45), (30, 0.40), (50, 0.35)
	fNIRS-PreT	(1, 0.45), (30, 0.40), (50, 0.35)

[1] The "Flooding" format is (epoch, flooding level).

TABLE VI
AVERAGE ACCURACY OF THE TEST SETS

Method	Dataset A	Dataset B	Dataset C
k-NN	69.20 $\pm$ 4.39	57.39 $\pm$ 2.99	41.61 $\pm$ 1.59
ANN	75.79 $\pm$ 4.32	58.66 $\pm$ 2.38	47.54 $\pm$ 2.07
SVM	73.28 $\pm$ 5.93	60.99 $\pm$ 2.94	48.08 $\pm$ 2.17
CNN	87.00 $\pm$ 3.32	74.14 $\pm$ 1.78	72.89 $\pm$ 1.70
LSTM	88.45 $\pm$ 3.41	63.75 $\pm$ 1.65	61.94 $\pm$ 1.95
fNIRS-T	**95.29 $\pm$ 3.17**	76.59 $\pm$ 1.86	**75.49 $\pm$ 2.07**
fNIRS-PreT	94.84 $\pm$ 2.71	**78.53 $\pm$ 1.83**	70.66 $\pm$ 1.45

The bold indicates the best result on the test set.

C. Classification Results

Table VI summarizes the average accuracy (mean $\pm$ std %) of the test sets. Fig. 5 shows the classification accuracy of individual subjects for LOSO-CV, and the relevant results are listed in Table VII. In Table VI, the classification performance of fNIRS-T is far superior to the baseline methods (Wilcoxon signed-rank test, $p < 0.001$). For the traditional classifiers, the classification results of ANN and SVM are better than k-NN.

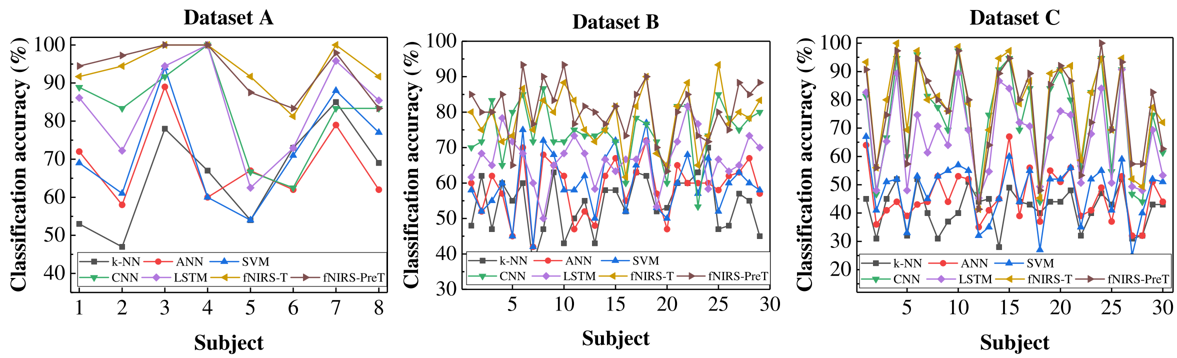


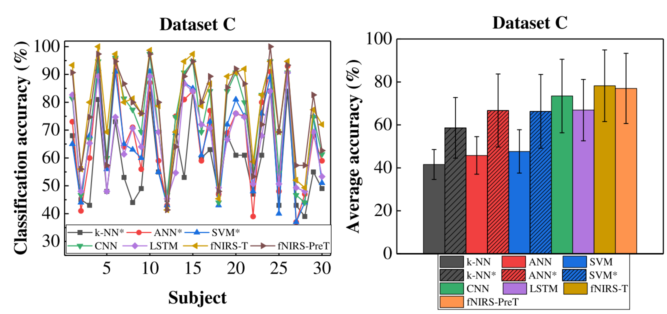
Fig. 5. Classification accuracy for individual subjects on the three datasets.

TABLE VII
AVERAGE CLASSIFICATION RESULTS OF ALL INDIVIDUAL SUBJECTS

Dataset	Method	Accuracy	Precision	Recall	F1-score	Kappa
Dataset A	k-NN	65.71 ± 12.44	70.00 ± 15.98	58.68 ± 10.78	63.40	0.31
	ANN	68.84 ± 9.92	70.18 ± 12.05	67.88 ± 8.74	68.73	0.38
	SVM	71.88 ± 12.98	72.74 ± 14.01	70.31 ± 15.04	71.23	0.44
	CNN	82.47 ± 11.64	83.96 ± 13.59	82.29 ± 10.66	82.70	0.65
	LSTM	83.68 ± 12.42	84.88 ± 12.62	82.99 ± 13.41	83.55	0.67
	fNIRS-T	93.84 ± 5.99	96.30 ± 4.22	91.15 ± 10.09	93.42	0.88
	fNIRS-PreT	92.97 ± 6.70	93.55 ± 7.39	93.06 ± 10.58	92.79	0.86
Dataset B	k-NN	54.25 ± 7.63	53.75 ± 6.48	66.78 ± 10.30	59.21	0.09
	ANN	58.68 ± 7.61	59.25 ± 8.71	62.30 ± 12.13	59.80	0.17
	SVM	59.83 ± 8.51	60.40 ± 9.37	61.03 ± 9.47	60.28	0.20
	CNN	74.43 ± 7.63	77.65 ± 10.59	72.30 ± 13.95	73.41	0.49
	LSTM	66.78 ± 6.81	69.53 ± 9.51	63.45 ± 11.09	65.35	0.34
	fNIRS-T	78.16 ± 7.59	79.49 ± 8.35	76.90 ± 11.97	77.59	0.56
	fNIRS-PreT	80.57 ± 7.53	80.91 ± 9.66	82.30 ± 12.13	80.67	0.61
Dataset C	k-NN	41.56 ± 6.95	41.97 ± 7.23	41.56 ± 6.95	40.95	0.12
	ANN	45.78 ± 8.76	46.88 ± 9.05	45.78 ± 8.76	45.21	0.19
	SVM	47.64 ± 10.10	48.38 ± 10.48	47.64 ± 10.10	47.38	0.21
	CNN	73.47 ± 17.11	75.34 ± 16.07	73.47 ± 17.11	72.73	0.60
	LSTM	66.89 ± 14.29	67.92 ± 14.60	66.89 ± 14.29	66.19	0.50
	fNIRS-T	78.22 ± 16.69	79.36 ± 16.67	78.22 ± 16.69	77.59	0.67
	fNIRS-PreT	76.98 ± 16.35	77.66 ± 16.09	76.98 ± 16.35	76.72	0.65

Due to the powerful learning capabilities of CNN and LSTM, they are significantly better than traditional classifiers (Wilcoxon signed-rank test, $p < 0.001$). For the smallest Dataset A, fNIRS-T, CNN, and LSTM achieve an average accuracy of 95.29%, 87.00%, and 88.45%, respectively. On the largest Dataset C, these models obtain 75.49%, 72.89%, and 61.94%, respectively. In the individual subject classification experiment, the proposed models not only achieve excellent performance but also have significant differences (paired t-test, $p < 0.001$). Specifically, fNIRS-T obtains an average subject accuracy of 78.22% and surpasses CNN and LSTM by a large margin of +4.75% and +11.33% on Dataset C. The classification accuracy of fNIRS-PreT also exceeds 80% on Dataset B. These indicate that our networks have a strong generalization ability to overcome individual differences. F1-score and Kappa coefficient further illustrate this point. It is worth noting that the classification performance of LSTM is worse than CNN. One possible reason is that fNIRS signals of many inactive brain regions have no obvious sequence patterns.

The classification performance of traditional classifiers relies on hand-crafted features. To explain this point, the well-designed features in [26] are used for classification on Dataset C. The signals of three-time windows in the ranges of 0–5 s, 5–10 s,



(a) (b)

Fig. 6. (a) shows the classification accuracy of individual subjects. (b) presents the comparison results of test sets. The superscript * indicates well-designed features.

10–15 s within epochs are extracted to calculate the average of each channel [26]. Fig. 6(a) shows the classification accuracy of individual subjects. Fig. 6(b) presents the comparison results of test sets. Compared with the statistical features, the well-designed features can greatly improve classification accuracy (Wilcoxon signed-rank test, $p < 0.001$). The average accuracy of test sets for k-NN, ANN, and SVM achieve 59.34 ± 2.26%

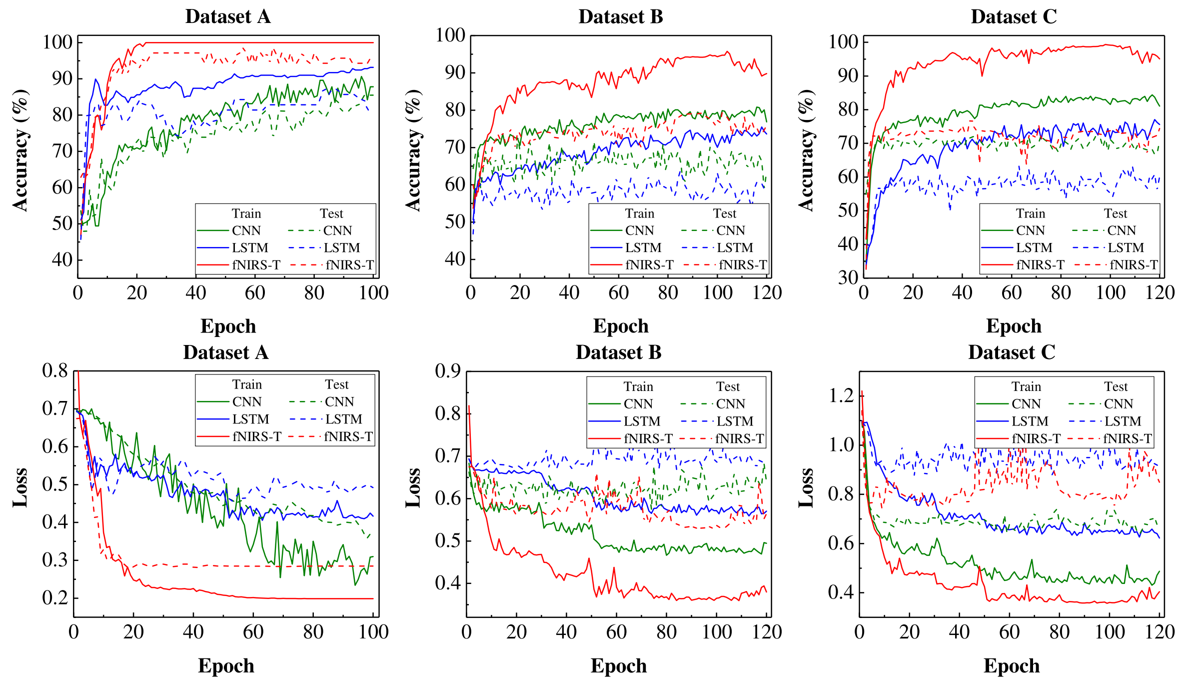


Fig. 7. The training and test curves of CNN, LSTM, and fNIRS-T. Top row: the curves of training and test accuracy. Bottom row: the curves of training and test loss. Under the same training epochs, fNIRS-T is stronger than the baseline models for convergence speed and trainability.

(+17.73%), 68.70 ± 2.24% (+21.16%), and 67.76 ± 2.07% (+19.68%), respectively. Nevertheless, these traditional classifiers are at least 6.79% lower than fNIRS-T. These classifiers have reached the classification bottleneck and it is difficult to continue to improve the classification accuracy. In addition, our models also have a lower standard deviation. Therefore, the proposed models can significantly enhance the BCI performance.

D. Training and Test Curves

The training and test curves of CNN, LSTM, and fNIRS-T are shown in Fig. 7. fNIRS-T has higher accuracy compared with other deep learning methods. The training loss curve converges steadily around the final flooding level, and the test loss curve does not increase. The training accuracy continues to increase and the test accuracy does not decrease. Note that the short-term accuracy and loss degradation of fNIRS-T is caused by the restart phase of cosine learning rates, but it is beneficial in the long term. Flooding prevents the performance deterioration caused by overfitting and flooding levels of different models may be different. Since the training loss is restricted around the flooding level, these training loss curves may not provide a rigorous comparison, but fNIRS-T usually achieves lower training and test loss. Furthermore, under the same training epochs, fNIRS-T is stronger than the baseline models for convergence speed and trainability.

E. fNIRS-PreT Performance

We first demonstrate the role of the preprocessing module, and then comparison experiments verify the effectiveness of fNIRS-PreT. The signal processing flow of the preprocessing module is shown in Fig. 8. The Raw signal contains instrument noise, physiological noise, and motion artifacts. The first 1D average pooling layer $AvgPool\_1$ with a smaller kernel retains the main information of the raw signal and removes signal spikes. After that, the signal is smoothed by $AvgPool\_2$ with a larger kernel and is finely smoothed by $AvgPool\_3$. Finally, the signal is scaled and shifted by layer normalization. Average pooling and layer normalization have the functions of filtering and baseline correction, respectively. Moreover, layer normalization can also accelerate network training and convergence. The experimental results are summarized in Table VIII. First, data preprocessing

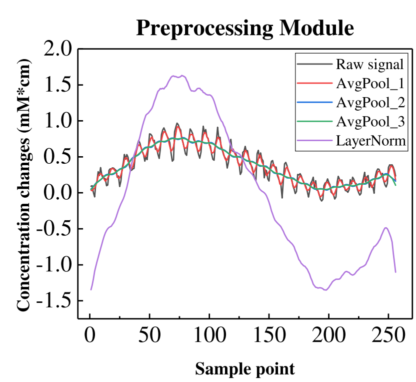


Fig. 8. Signal processing flow of the preprocessing module.

TABLE VIII
Average Accuracy of the Test Sets

Model	Preprocessing	Dataset A	Dataset B	Dataset C
fNIRS-T	×	91.15 ± 2.57	71.44 ± 2.59	65.60 ± 2.33
fNIRS-T	√	95.29 ± 3.17	76.59 ± 1.86	75.49 ± 2.07
fNIRS-PreT	×	94.84 ± 2.71	78.53 ± 1.83	70.66 ± 1.45

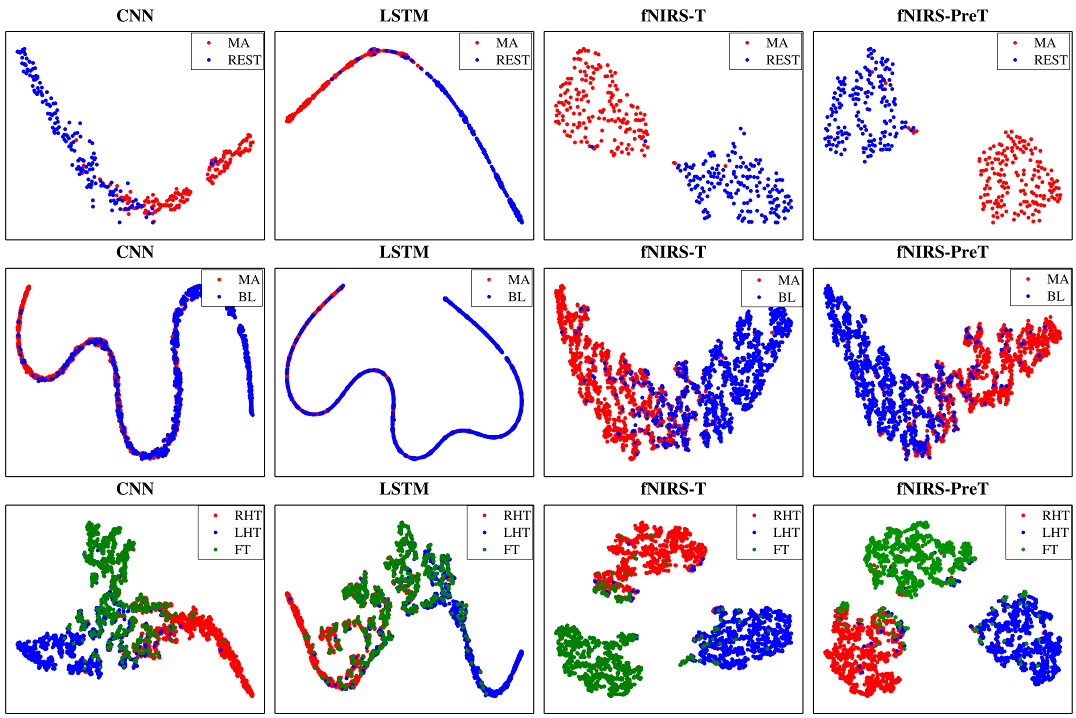


Fig. 9. The top row is the visualization result of Dataset A, the middle row is Dataset B, and the bottom row is Dataset C.

is essential for fNIRS classification tasks. The classification accuracy decreases when filtering and baseline correction are removed. The accuracy of fNIRS-T decreased by 4.14%, 5.15%, and 9.89%, respectively. Second, fNIRS-PreT using raw signals has excellent classification performance. On Dataset A, fNIRS-PreT is competitive to fNIRS-T and the differences are not significant (Wilcoxon signed-rank test, $p = 0.654$). It is slightly lower than fNIRS-T on Dataset C but it is 1.94% higher on Dataset B. Therefore, the role of the preprocessing module is proven and makes fNIRS-PreT a full end-to-end classification network.

V. Discussion

A. Visualization of Feature Maps

To further demonstrate the classification performance of the proposed models, t-SNE [43] is used to visualize the features learned by models in the 2D embedding space. As shown in Fig. 9, the features learned by fNIRS-T and fNIRS-PreT have higher separability. Our proposed models have similar feature distributions, and the three classes are divided into three highly separable clusters on Dataset C. These intra-clusters are much tighter, and inter-clusters separate in a regular triangular structure. However, CNN and LSTM do not have such obvious feature distribution. The visualization results also confirm that our models have stronger learning abilities and obtain the best classification performance in Table VI and Table VII.

B. Ablation Study

To verify the effect of the spatial-level and channel-level representation, we remove them deliberately for ablation experiments. Table IX shows the average accuracy of test sets. Compared with the ablation networks (fNIRS-ST and fNIRS-CT), fNIRS-T has higher accuracy. Any part of the network is removed will cause performance degradation. Fig. 10 records the curves of training and test. Since these networks are trained under the same flooding levels, they have the same level of training loss. However, fNIRS-T has lower test loss compared with the ablation networks. Note that the periodic restart of cosine learning rates causes periodic fluctuations in the accuracy and loss curves. The spatial-level and channel-level representation are essential for classification and have practical significance in brain imaging. In the MA task, it is confirmed that a relative focal bilateral increase of HbO in the dorsolateral prefrontal cortex (DLPFC), while a decrease in the medial area of the anterior prefrontal cortex (APFC) at the same time [23], [44], [45]. The HbO response is significant in the left DLPFC and APFC, but the HbR response is smaller and insignificant [23]. The spatial-level representation can cover multiple channels to detect DLPFC and APFC, and the channel-level representation can further focus on the HbO and HbR response of a single channel.

TABLE IX
Average Accuracy of the Test Sets

Model	Dataset A	Dataset B	Dataset C
fNIRS-ST	93.45 $\pm$ 2.85	75.92 $\pm$ 1.85	74.79 $\pm$ 1.77
fNIRS-CT	94.31 $\pm$ 2.25	76.10 $\pm$ 1.62	74.62 $\pm$ 1.48
fNIRS-T	**95.29** $\pm$ **3.17**	**76.59** $\pm$ **1.86**	**75.49** $\pm$ **2.07**

Furthermore, we explore the role of flooding on Dataset C. In Fig. 11, flooding can significantly improve test accuracy and reduce test loss by keeping training loss around the flooding level. Not to be overlooked, flooding also prevents baseline models

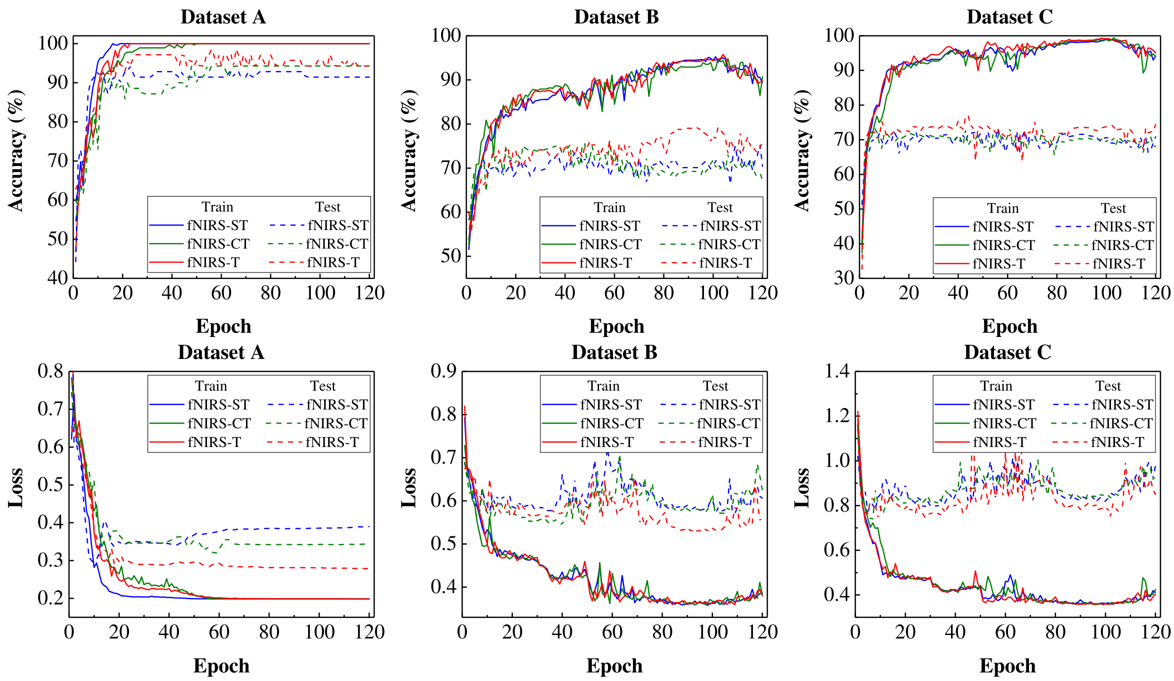


Fig. 10. These curves show that the training and test process of fNIRS-T and ablation networks (fNIRS-ST and fNIRS-CT).

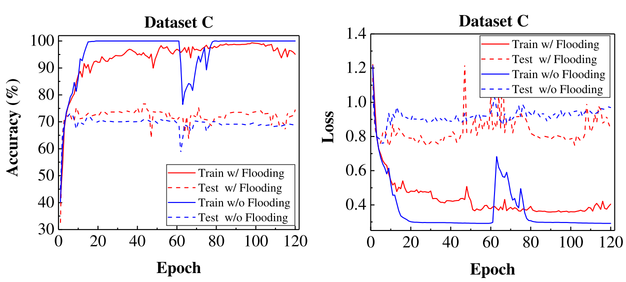


Fig. 11. The learning curves of fNIRS-T. Flooding can improve test accuracy and alleviate the impact of cosine learning rates in the restart phase.

from overfitting, which potentially enhances the classification results. The real performance gap between baseline networks and our models will be further expanded. In addition, flooding can also alleviate the impact of deteriorating performance caused by restarting cosine learning rates. Therefore, it is an essential technique for training models efficiently.

C. Limitation

Although fNIRS-T and fNIRS-PreT have excellent classification performance, many hyperparameters need to be tuned. The inherent problems of Transformers cannot be ignored. The memory and computational complexity of self-attention are quadratic in the input sequence length [10]. The huge amount of parameters brings the risk of overfitting. These factors may limit the BCI application. Flooding is a novel method to solve overfitting. In experiments, we find that it is more effective than other methods, such as dropout and early stopping, but it requires multiple tests to choose flooding levels.

VI. Conclusion

The fNIRS classification problem has always been the focus of the BCI field. In this work, we analyze the spatial-level and channel-level representation of fNIRS signals and propose an fNIRS classification network based on Transformer, called fNIRS-T. Moreover, we propose an end-to-end classification network named fNIRS-PreT by using a preprocessing module, which is composed of 1D average pooling and layer normalization to replace filtering and baseline correction. Since the size of fNIRS datasets is usually small, label smoothing and flooding are applied to improve the generalization ability of models. Comprehensive experiments based on three open-access datasets are conducted to validate the effectiveness of the proposed models. Ablation studies prove the importance of spatial-level and channel-level representation. The t-SNE visualization further demonstrates that our models have excellent feature learning abilities.

In future work, we will explore more efficient fNIRS representation methods and incorporate more inductive biases into Transformers to study instant classification [12]. Due to the preciousness of fNIRS data, this may limit the application of deep networks. Data augmentation may be a potential solution.

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